AJAY ELIKA

Software Engineer | +1(913)762-3528 | ajayelika0550@gmail.com

PROFESSIONAL SUMMARY

Full-Stack Software Engineer and AI Enthusiast with a strong foundation in **scalable systems, cloud computing, and machine learning**. Passionate about **deep learning, large language models (LLMs), and research-driven innovation**. Proven expertise in **end-to-end system design, RESTful APIs, and cloud-native architectures** (Azure/AWS). Actively contributes to open-source projects, including **OpenClaw** (approved pull request, listed as contributor). Seeking to leverage technical and research skills in a **master's program focused on AI/ML and advanced data science**.

Profile Links:

- Portifolio-Website: https://www.ajayelika.cc/
- GitHub: https://github.com/ajay99511
- LinkedIn: https://www.linkedin.com/in/e-aj-47b71238b/
- LeetCode: https://leetcode.com/u/ajay216/
- Explore the UI demonstration of my github projects at https://www.ajayelika.cc/projects

EDUCATION

University of Central Missouri, Warrensburg, MO | GPA 3.5
Master of Science in Computer Science | Dec 2022 - May 2024

Jawaharlal Nehru Institute of Technology, Hyderabad, India | GPA 3.13
Bachelor of Technology in Computer Science | Aug 2018 - May 2022

CERTIFICATIONS & ACHIEVEMENTS

- **Microsoft Certified: Azure Developer Associate (AZ-204)** | Microsoft
- **Technical Support Fundamentals** | Google (via Coursera) | 2023
- **Python for Everybody Specialization** | University of Michigan (via Coursera) | 2022
- **Contributor to OpenClaw project (Approved Pull Request)**

TECHNICAL SKILLS

AI/ML & Data Science

- **Frameworks:** scikit-learn, TensorFlow/PyTorch (self-study), Pandas, NumPy, Seaborn, Matplotlib
- **Techniques:** Supervised/Unsupervised Learning, Ensemble Methods (Random Forest, SVM, KNN), Feature Engineering, Model Evaluation (Cross-Validation, Accuracy Metrics)

Full-Stack Development

- **Frontend:** React.js, React Native (Expo), Angular, Next.js, Redux, HTML5/CSS3, Bootstrap, Flutter (Dart)
- **Backend:** Node.js, Express.js, ASP.NET Core, .NET Core, Flask, Socket.io
- **Databases:** PostgreSQL, SQL Server, MySQL, Azure Cosmos DB, Redis (Caching), ObjectBox, Drift ORM, Room

Cloud & DevOps

- Azure (App Service, Functions, Blob Storage, Service Bus, AKS, Key Vault), AWS, Docker, Kubernetes, CI/CD (Azure DevOps/GitHub Actions)

APIs & Integration

- RESTful APIs, GraphQL, Stripe, FedEx API, OAuth 2.0, Passport.js, SignalR

Tools & Practices

- Git, Jest, NUnit, TDD, Agile/Scrum, Microservices, Event-Driven Architecture, JWT Authentication

WORK EXPERIENCE

Software Engineer | SLK Holdings | Jul 2024 – Present
Client: AZDOR (AZTaxes Application) | Jan 2026 – Present

- Developed interactive frontend modules using React, implementing batch-processing workflows and integrated image viewers to streamline the entry of Arizona state tax forms for high-volume operations.
- Built and maintained RESTful API endpoints using .NET, implementing Role-Based Access Control (RBAC) to enforce secure data access layers for Operator, Admin, and Supervisor personas.
- Optimized PostgreSQL database performance by designing normalized schemas and efficient queries for managing document hierarchies, ensuring data integrity across the CRUD lifecycle.
- Streamlined document submission workflows by implementing state-driven logic to manage the transition of batches from initial data entry to Supervisor review, reducing processing errors.
- Collaborated within an Azure-hosted environment, utilizing Google Workspace for technical documentation and sprint coordination to ensure timely delivery of internal features.

Client: Castoffs (C2C Marketplace) | July 2024 – Dec 2025

- Developed a full-stack C2C marketplace using Node.js and React Native (Expo), integrating Stripe to facilitate secure peer-to-peer transactions and real-time messaging between users.
- Designed RESTful APIs and server-side routes for authentication and product management, implementing optimized data retrieval patterns that reduced page load times by 40%.

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- Configured cloud-native infrastructure on Azure, utilizing App Services and Cosmos DB with Kubernetes orchestration to support scalable mobile and web environments.
- Implemented event-driven pipelines using Azure Service Bus and BullMQ for order fulfillment and automated notifications, reducing manual intervention in the transaction lifecycle by 90%.
- Optimized PostgreSQL schemas and implemented Redis caching for shipping rates and third-party tokens, cutting redundant API calls by 75% and reducing operational costs.

GRADUATE PROJECTS

Full-Stack Dating Application | University of Central Missouri

Advanced Database Systems (CS 5600) | Sep 2023 - May 2024

GitHub: https://github.com/ajay99511/DatingApp216

Tech Stack: .NET 8, Angular 16, SQL Server 2022, Entity Framework, SignalR, ADO.NET

- Developed a real-time messaging platform using .NET 8 and SignalR, achieving low-latency (<500ms) instant communication and push notifications for 100+ concurrent users.
- Architected a secure backend with Identity Framework and JWT authentication, enforcing Role-Based Access Control (RBAC) to protect user data and communication channels.
- Designed a normalized SQL Server database using Entity Framework Core, optimizing relational queries and stored procedures to reduce API response times by 35%.
- Created a modular Angular frontend with reusable components and interceptors, ensuring a responsive UI and secure navigation via custom route guards.

Diabetes Prediction Using Machine Learning | University of Central Missouri

Machine Learning (CS 5710) | Jan 2023 - May 2023

Github: https://github.com/ajay99511/Diabetes-Prediction

Tech Stack: Python, scikit-learn, Flask, Pandas, NumPy, Seaborn, Matplotlib

- Developed an ensemble learning model (Random Forest, SVM, KNN) that achieved 87% prediction accuracy, outperforming baseline models by 12% through hyperparameter tuning.
- Automated a data preprocessing pipeline for feature engineering and standardization of 1,000+ patient records, improving model interpretability and stakeholder insights.
- Deployed a real-time prediction API using Flask, delivering results with <200ms latency and integrating model storage with local file systems for rapid deployment.
- Evaluated model performance through rigorous cross-validation and feature importance analysis, identifying key risk factors like glucose levels and BMI-to-age ratios.

Soil Moisture Prediction Using Random Forest | Undergraduate Final Project

Vasireddy Venkatadri Institute of Technology, India | May 2021 - Mar 2022

Github: https://github.com/ajay99511/Soil-Moisture-Prediction

Tech Stack: Python, scikit-learn, Pandas, NumPy, SciPy, Matplotlib

- Built a Random Forest Classifier to predict groundwater availability with 82% accuracy, utilizing environmental features to address real-world agricultural challenges.
- Performed statistical data preprocessing on 5,000+ data points using SciPy and Pandas, resolving missing values and normalizing features to ensure high-quality model inputs.
- Implemented feature scaling and transformation techniques to standardize inputs, significantly improving algorithm convergence and model robustness during training.
- Optimized the predictive model through systematic hyperparameter tuning in scikit-learn, demonstrating the practical application of ML to environmental resource management.

Open-Source Personal Projects (GitHub):

Deep Learning Projects

GitHub: https://github.com/ajay99511/DL_Algorithms

- Engineered a decoder-only GPT-style Transformer and Vision Transformer (ViT) from scratch using PyTorch, implementing custom multi-head causal attention, fused QKV projections, and PatchEmbedding layers.
- Implemented an RLHF (Reinforcement Learning from Human Feedback) pipeline featuring a REINFORCE-style policy gradient with KL-divergence penalties to align model outputs with reward functions.
- Optimized autoregressive inference latency by designing a KV-Caching system that eliminated redundant computations, improving generation speed for long-context sequences.
- Developed a diagnostic suite for neural network health, utilizing custom PyTorch hooks to monitor activation statistics,

FastBeat — Kotlin · Jetpack Compose · Media3 ExoPlayer · Room · Hilt

GitHub: https://github.com/ajay99511/FastBeat

- Built a 14-screen Android app (API 35, backward-compatible to Android 8.0) using **100% Jetpack Compose (Material 3)**, **Media3 ExoPlayer**, and **Room DB** (8 entities, 30+ DAOs) for playlists, bookmarks, analytics, and a persistent now-playing queue.

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- Implemented **MVVM with Hilt/Dagger DI** (3 scopes), 5 domain-specific ViewModels, and a **Kotlin Coroutines Flow pipeline** (StateFlow, combine, flatMapLatest) for strict layer separation and zero LiveData usage.
- Fullscreen video player with **swipe gestures** (volume/brightness/seek), **Picture-in-Picture (Android 8+)**, analytics dashboard (playtime streaks, continue-watching), **Android 11+ Scoped Storage compliance**, and **background thumbnail generation** in isolated SupervisorJob to prevent UI blocking.

DayVault (Memory Palace) — Flutter · Dart · ObjectBox · Riverpod · AES-256
GitHub: ***https://github.com/ajay99511/DayVault***

- Developed for **Android/iOS/Desktop** using **Flutter + Riverpod 3.x**, with a **Service-Repository pattern**, **ObjectBox NoSQL**, and **Freezed** for immutable, code-generated models.
- Implemented **PBKDF2 key derivation (HMAC-SHA256, 100K iterations)**, **AES-256-CBC field-level encryption** (version-prefixed ciphertext), **biometric unlock**, and **isolate-based crypto** to prevent UI thread blocking.
- Glassmorphic design with **animated orbs**, **calendar heatmap journaling**, and **drag-to-reorder preference tracking**, applying **10+ design patterns** (Repository, Strategy, Chain of Responsibility).

Chronos Planner (Chronosky) — Flutter · Dart · Drift ORM · SQLite · Provider
GitHub: ***https://github.com/ajay99511/chronos_planner***

- Desktop-first (Windows/macOS/Linux/iOS/Android) with **Flutter + MVVM + Repository Pattern**, abstracting **Drift ORM/SQLite** for future cloud-sync integration with **zero UI refactoring**.
- Used **Drift .watch() streams**, **Provider (ChangeNotifier)**, and **optimistic UI updates** with a **5-version schema migration** (including SharedPreferences-to-SQLite migration) for zero-latency sync.
- **Glassmorphic design** (7 screens, 8 custom widgets), **Focus Mode HUD** (always-on-top window), **algorithmic productivity analytics** (energy peak detection, task ROI), and **undo/redo stack** with task snapshot recovery.

MD Explorer — Flutter · Dart · Riverpod · GFM · Offline-First
GitHub: ***https://github.com/ajay99511/md_doc_viewer***

- Built for **Windows/macOS/Linux/Android/iOS** with **Flutter + Riverpod**, supporting **full GFM rendering**, **syntax highlighting (20+ languages)**, and a **custom stylesheet**.
- Implemented **lazy-loading file tree** with **LRU cache (500 entries, 60s TTL)**, **real-time directory watching** (broadcast streams, path-prefix cache invalidation), and **instant startup** regardless of directory size.
- **3-tier layout system** (Mobile/Tablet/Desktop), **Android Scoped Storage compliance**, and **auto-dispose memory management** (FutureProvider.autoDispose).

RESEARCH INTERESTS

- **Deep Learning:** Neural network architectures, transfer learning, and optimization techniques.
- **Large Language Models (LLMs):** Pre-training, fine-tuning, inference optimization, and prompt engineering.
- **Cloud-AI Integration:** Scalable ML deployments, edge AI, and model serving (Azure ML/AWS SageMaker).